Ain Shams University

Faculty of Computer and Information Sciences

Statistical Analysis of Second-Hand Smoke Exposure:

A Comprehensive Study

Made By:

Ahmed Yasser Hussein Sayed - 20241701501

Yussuf Eslam Sanaa Abdelsalam - 20241701850

Omar Ahmed Karem Mahmoud – 20241701505

Supervised By:

Dr. Ghada Hamed

Dr. Mohamed Gasser

Introduction

Second-hand smoke, commonly referred to as passive smoking, is the involuntary inhalation of tobacco smoke from the surrounding environment. It contains many harmful substances known to have adverse effects on human health. Despite public health efforts aimed at reducing smoking, exposure to second-hand smoke remains prevalent, mainly at home and in social settings.

This project examines the relationship between second-hand smoke exposure and various health and behavioral outcomes, including headache frequency, respiratory symptoms, smoking initiation, physical activity, and stress levels. Using a dataset of 100 individuals, the study applies descriptive statistics, inferential hypothesis testing, and regression analysis to evaluate these relationships.

The aim of this analysis is to quantify the impact of second-hand smoke exposure using appropriate statistical methods and visualizations, and to highlight its potential health implications.

Data Description & Preprocessing

This section provides a detailed explanation of the dataset used for the analysis, along with all data loading, cleaning, and preprocessing steps executed in the R environment.

Dataset Overview

Attribute	Description
Dataset File	second_hand_smoke_dataset_100.csv
Final Dataset Size	100 observations (rows)
Output	Final dataset size: 100 rows

Variable Descriptions

The dataset comprises 100 observations across eight variables, categorized as follows:

Categorical / Factor Variables

Variable Name	Description	Levels (Set in R)
Exposed	Participant exposure to second-hand smoke.	"No", "Yes"
Started_smoking	Whether the participant has started smoking.	"No", "Yes"
Parent_smokes	Whether a parent smokes.	"No", "Yes"
Respiratory_symptoms	Severity of respiratory symptoms.	**Ordered Factor**: "None" < "Mild" < "Moderate" < "Severe"

Continuous / Numeric Variables

Variable Name	Description
Headaches_per_month	Count of headaches reported per month.
Hours_week_exposure	Hours per week of second-hand smoke exposure.
Physical_activity_hours_per_week	Hours per week of physical activity.
Stress_level_1_10	Self-reported stress level on a 1–10 scale.

Data Cleaning and Preprocessing Steps

Step	What was done	R Action	Output Produced
1. Loading the Data	The CSV file was read into R and stored in the data frame data.	read_csv	*N/A (Implicit in code execution)*
2. Report Start / Bookkeeping	Printed a header indicating the start of preprocessing.	cat	0. Data Preprocessing
3. Count Rows Before Deduplication	The initial number of rows was captured to calculate duplicates removed.	nrow	*N/A (Internal variable initial_rows)*
4. Duplicate Removal	Exact duplicate rows were removed to ensure all observations are unique.	distinct (from dplyr)	Number of duplicate rows removed: 0
5. Final Row Count	The final size of the processed dataset was confirmed.	nrow + cat	Final dataset size: 100 rows
6. Convert to Factors	Character columns were converted to the R data type factor. Respiratory_symptoms was explicitly set as an **ordered factor** to enforce the intended severity sequence.	factor (for all categorical variables); factor(..., ordered = TRUE) (for Respiratory_symptoms)	*N/A (Internal data structure change)*
7. Handling of Missing Values (NA)	No explicit imputation or removal of missing values was performed. Calculations in subsequent analysis steps (EDA, inferential tests) were performed using the na.rm = TRUE argument to ignore NA values.	Summary functions with na.rm = TRUE	*N/A (Output related to NA handling occurs during EDA)*
8. Handling Duplicate Values	Removed Duplicate values from the dataset if there were any.		**N/A**

Categorical Variable Frequencies

The frequencies of the categorical variables, which inform the class balance, are as follows:

Variable	Level	Count
Exposed	No	23
	Yes	77
Respiratory_symptoms	None	52
	Mild	26
	Moderate	14
	Severe	8
Started_smoking	No	84
	Yes	16
Parent_smokes	No	58
	Yes	42

Exploratory Data Analysis (EDA)

The Exploratory Data Analysis (EDA) section provides a summary of the data's characteristics through descriptive statistics and visualization interpretation.

R Statistical Commands Reference

Category	Function	Package	Purpose	Typical Output
Data Exploration	**mean(), median(), sd()**	**base R**	**Calculates descriptive statistics**	**Numerical summary statistics**
	Mode()	**DescTools**	**Finds the most frequent value**	**Mode value for each variable**
	IQR()	**base R**	**Calculates interquartile range**	**Spread measure resistant to outliers**
	table()	**base R**	**Creates frequency tables**	**Counts for categorical variables**
Statistical Tests	**t.test()**	**stats**	**Performs t-test (means comparison)**	**t-statistic, p-value, confidence intervals**
	kruskal.test()	**stats**	**Kruskal-Wallis non-parametric test**	**Chi-squared statistic and p-value**
	chisq.test()	**stats**	**Chi-square test of independence**	**Chi-square statistic and p-value**
Regression & Modeling	**lm()**	**stats**	**Fits linear regression model**	**Regression coefficients, R-squared, p-values**
	summary()	**stats**	**Summarizes model results**	**Detailed regression output**

Category	Function	Package	Purpose	Typical Output
	confint()	stats	Calculates confidence intervals	Lower and upper bounds for coefficients
	predict()	stats	Makes predictions from model	Predicted values with intervals
Data Manipulation	group_by()	dplyr	Groups data by variable	Grouped data frame for aggregation
	summarise()	dplyr	Calculates summary statistics	Summary statistics by group
Visualization	ggplot()	ggplot2	Initializes ggplot object	Plot object
	geom_histogram()	ggplot2	Creates histogram	Frequency distribution plot
	geom_boxplot()	ggplot2	Creates boxplot	Distribution comparison plot

Descriptive Statistics for Continuous Variables

The central tendency and dispersion for the four continuous variables are summarized below[1]:

Variable	Mean	Median	Mode	Standard Deviation (SD)	Interquartile Range (IQR)
Headaches_per_month	1.900	2.000	1	1.446	2.000
Hours_week_exposure	4.610	2.000	0	5.278	6.000
Physical_activity_hours_per_week	3.110	3.000	3	1.763	2.000
Stress_level_1_10	4.390	4.350	4	1.641	2.350

Frequency Tables for Categorical Variables

The following tables show the distribution of observations across the levels of the categorical factors:

Exposed	Count
No	23
Yes	77

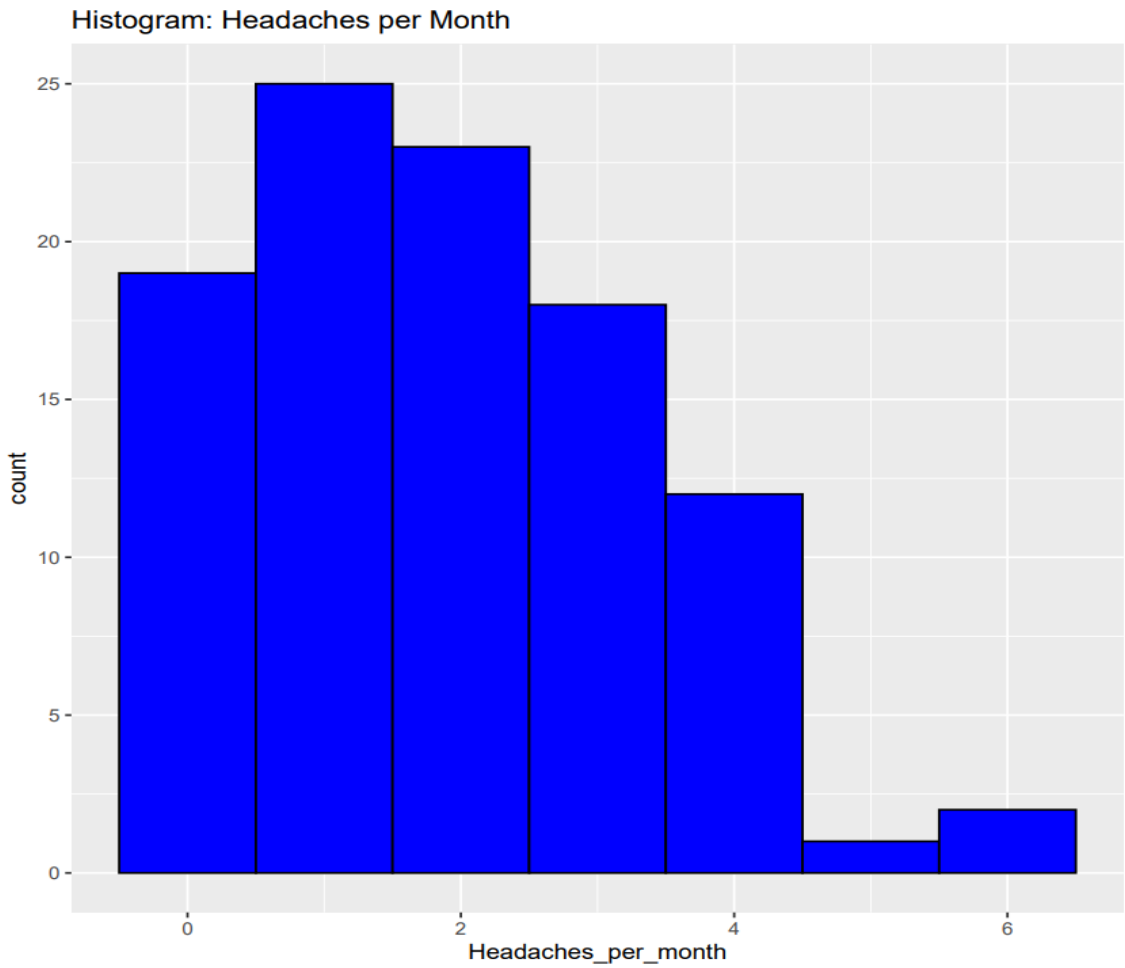
Respiratory_symptoms	Count
None	52
Mild	26
Moderate	14
Severe	8

Started_smoking	Count
No	84
Yes	16

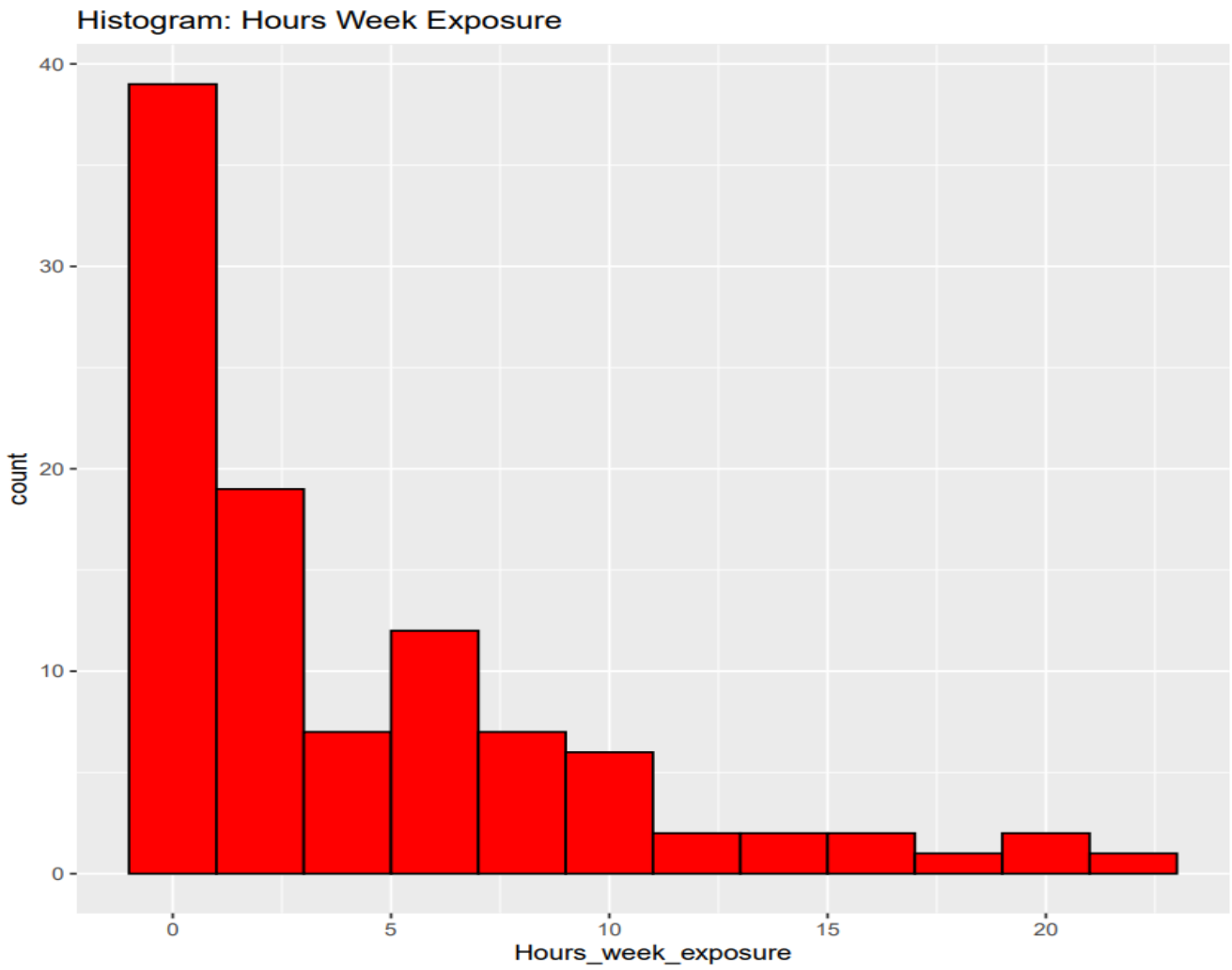
Parent_smokes	Count
No	58
Yes	42

Histograms

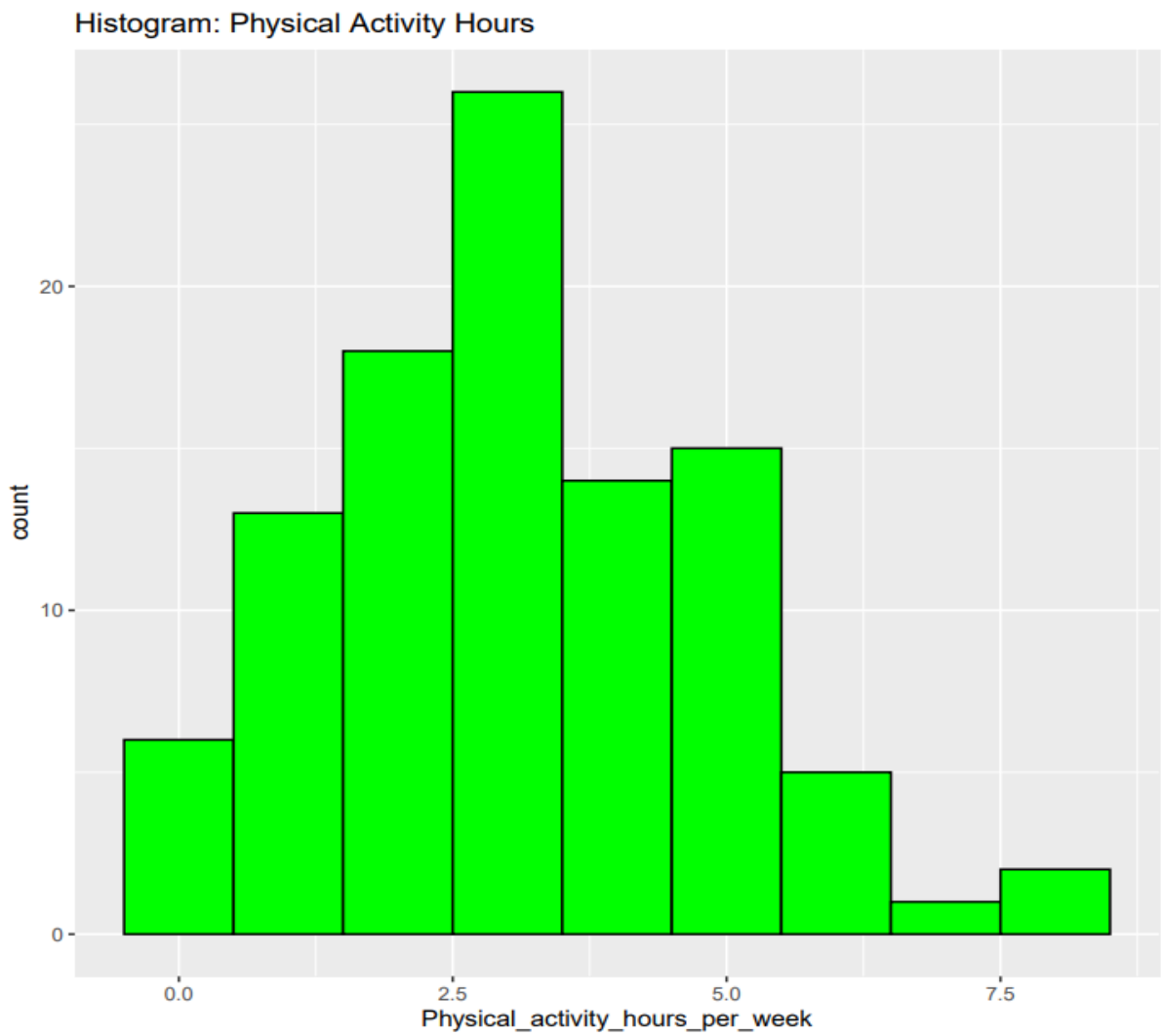
- **Headaches per Month:** The distribution is slightly skewed to the right, with a mode of 1 headache per month. The mean (1.9) and median (2.0) are very close, suggesting a modest degree of symmetry around the center



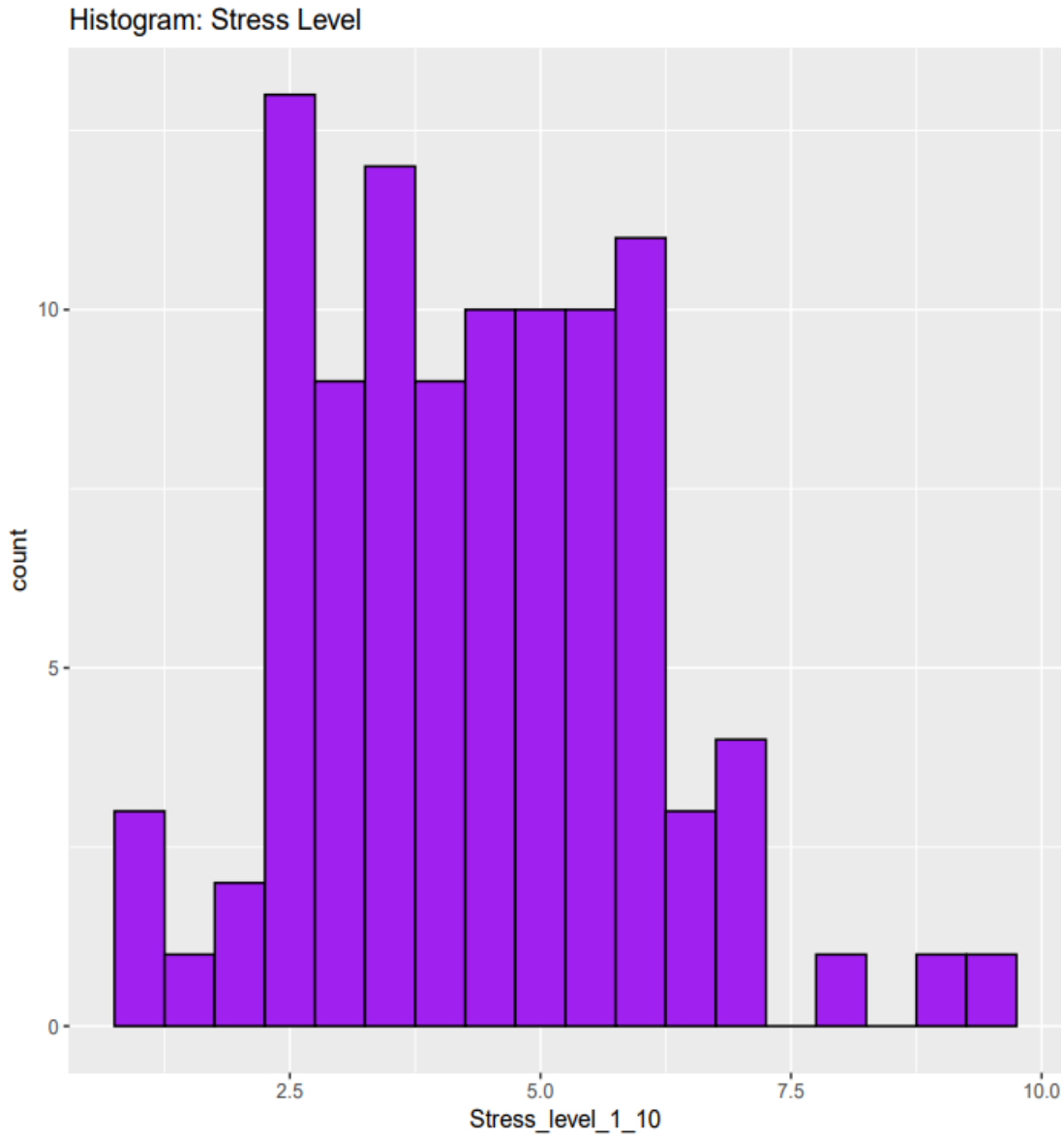
- **Hours Week Exposure:** This variable is highly right-skewed. The mode is 0 hours, and the median (2.0) is significantly lower than the mean (4.61), indicating a large concentration of observations at low values and a long tail of high exposure reports (outliers).



- **Physical Activity Hours per Week:** The distribution appears roughly unimodal and symmetric, with the mean (3.11) and median (3.0) being very similar. The highest frequency is centered around 3 hours per week.



- **Stress Level 1-10:** The distribution is spread out across the mid-range (2.5 to 6.5) and is approximately symmetric, with a slight peak at the mode of 4.



Results

This section presents the statistical results from the inferential analysis and linear regression, reported factually without interpretation.

Inferential Analysis: Hypothesis Test Statistics

Relationship	Test Used	Outcome	Group Means	Test Statistic	Degrees of Freedom (df)	p-value
R1: Headaches vs. Exposure	Welch Two Sample t-test	Headaches/Month	No: 0.957; Yes: 2.182	**t = -4.9432**	61.575	6.227 x 10^{-6}
R2a: Hours vs. Respiratory Symptoms	Kruskal-Wallis Rank Sum Test	Hours/Week Exposure	N/A	**x^2=22.017**	3	6.47 x 10^{-5}
R2b: Hours vs. Smoking Initiation	Welch Two Sample t-test	Hours/Week Exposure	No: 3.714; Yes: 9.313	**t = -3.2967**	17.877	0.004039
R3a: Physical Activity vs. Exposure	Welch Two Sample t-test	Physical Activity/Week	No: 3.174; Yes: 3.091	**t = 0.18936**	34.142	0.8509

Simple Linear Regression Results

Parameter	Coefficient Estimate	Standard Error	t value	p-value	95% Confidence Interval
Headaches_per_months	1.30975	0.17097	7.661	1.33 x 10^{-11}	[0.970, 1.649]
Hours_week_exposure	0.12804	0.02447	5.233	9.49 x 10^{-7}	[0.079, 0.177]

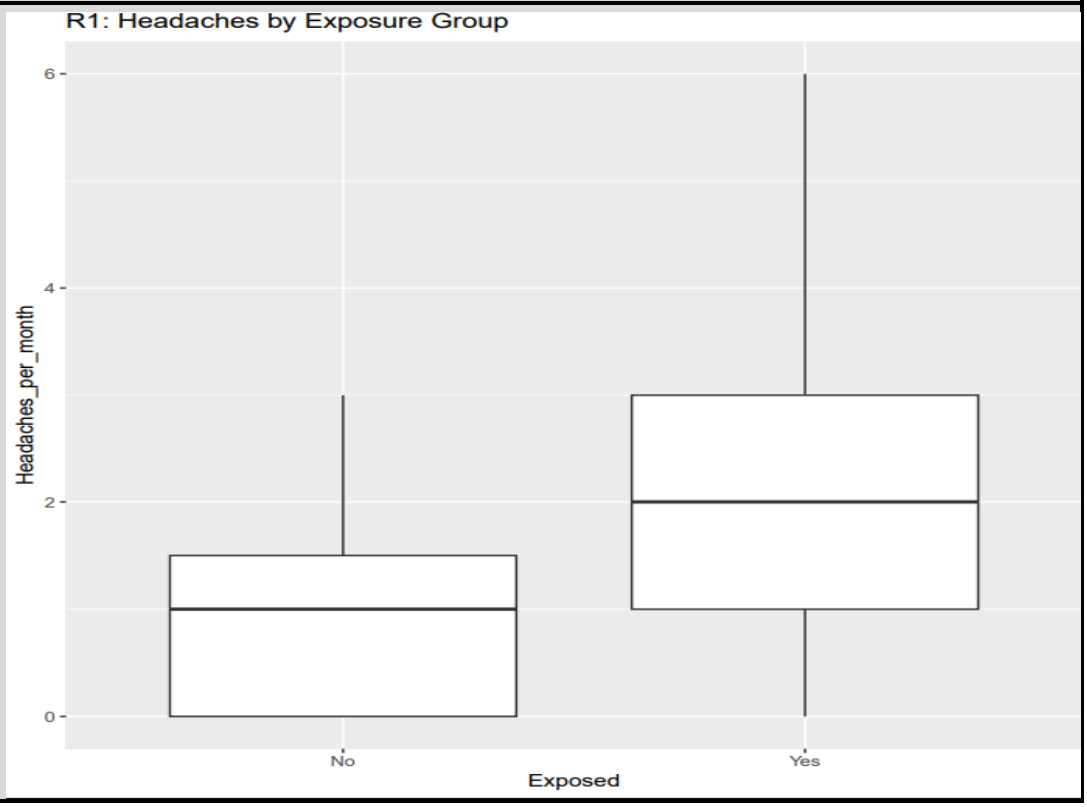
Prediction Intervals from Regression

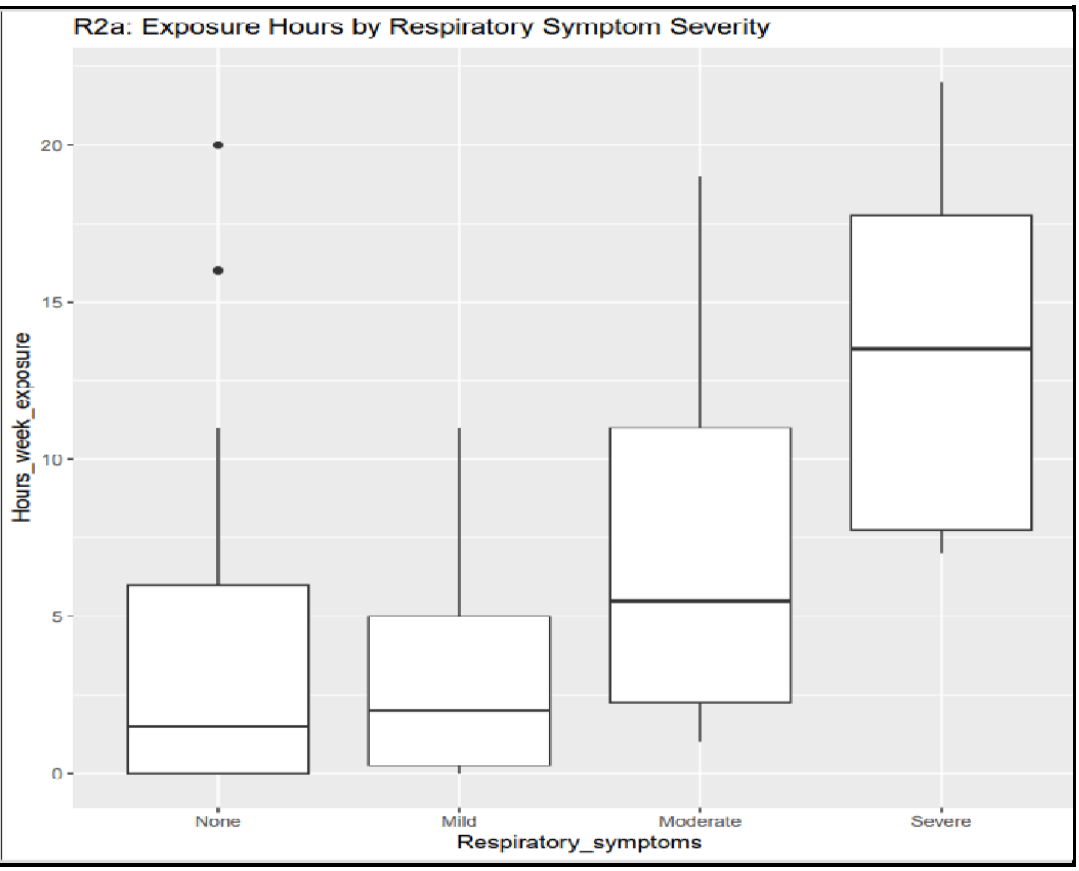
Hours_week_exposure	Predicted Headaches	95% Prediction Interval (lwr, upr)
0	1.30975	[-1.26253, 3.88203]
10	2.59012	[0.01427, 5.16597]

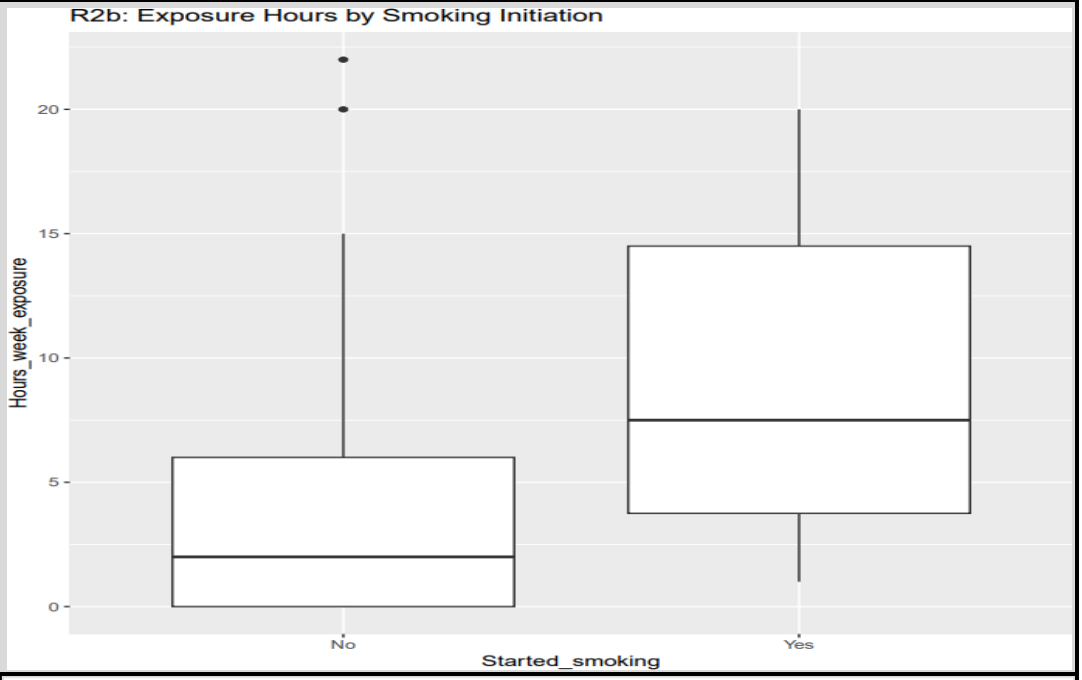
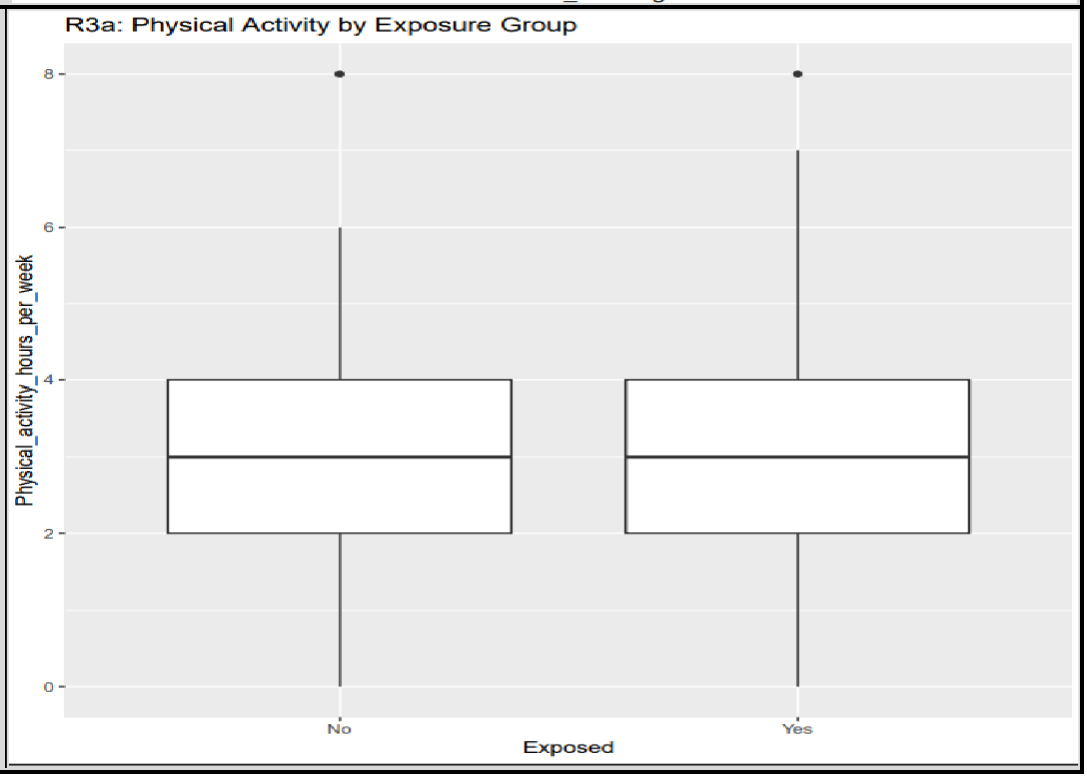
Population Mean Confidence Intervals

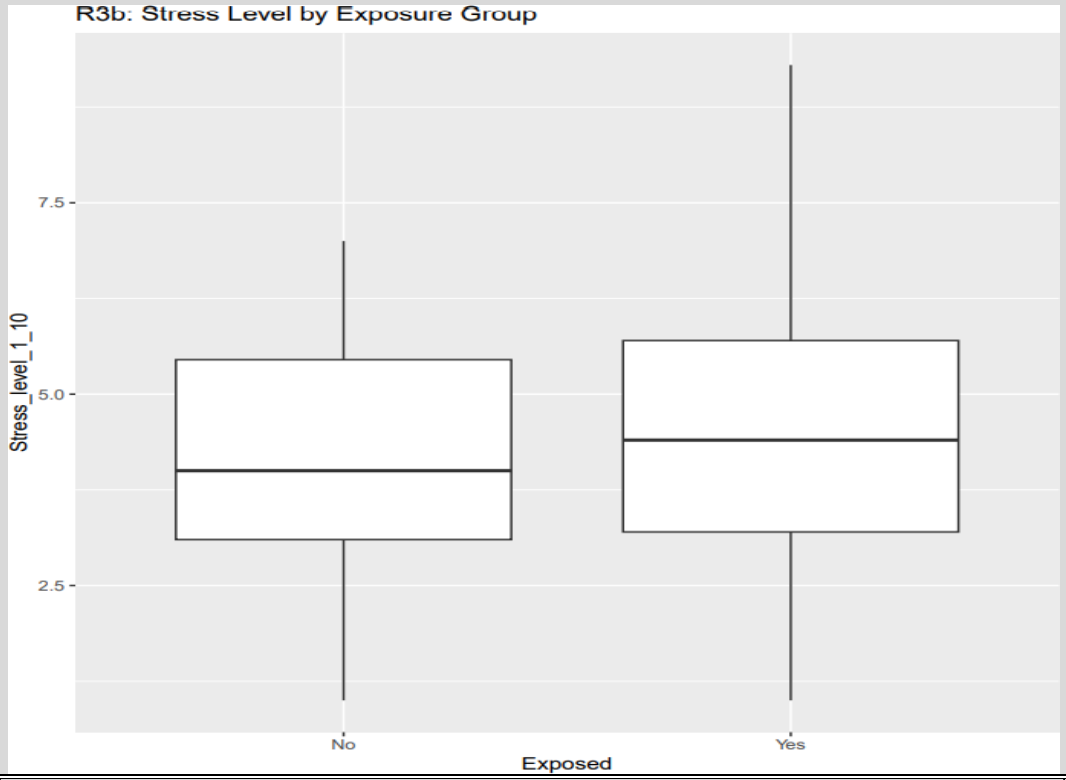
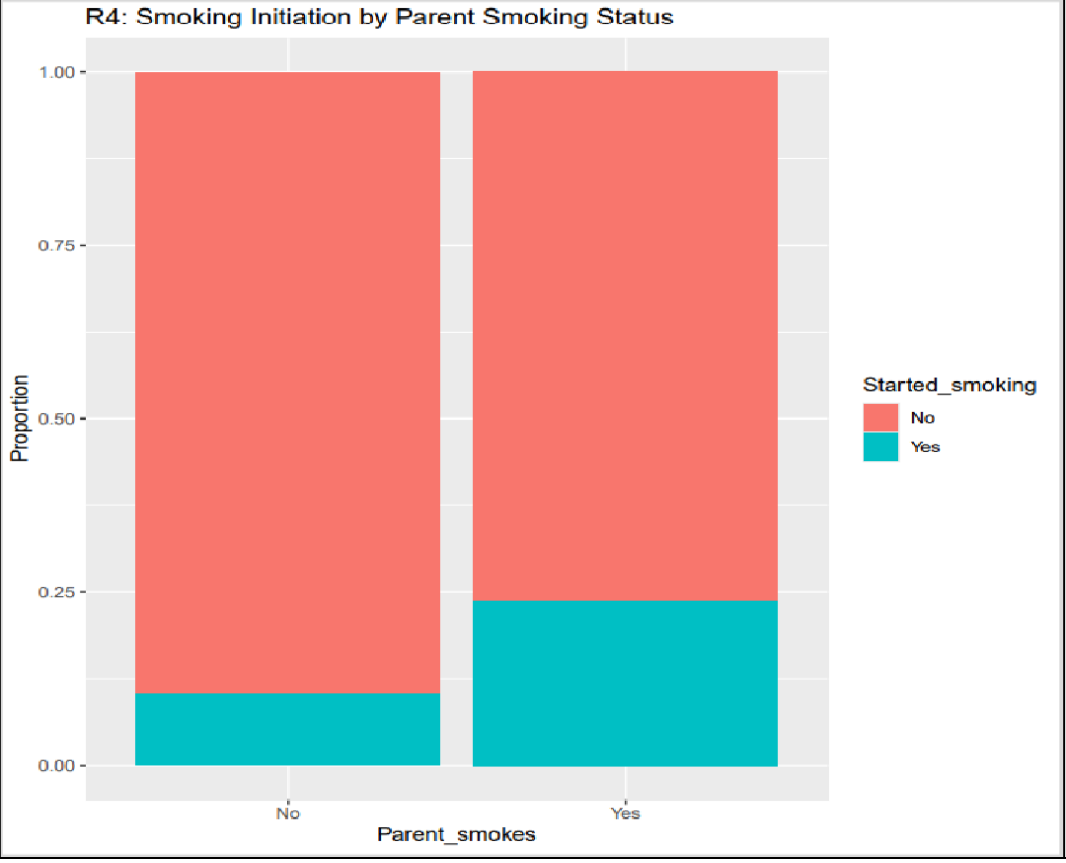
Variable	95% Confidence Interval for Mean (μ)
Headaches_per_month	[1.613, 2.187]
Hours_week_exposure	[3.563, 5.657]

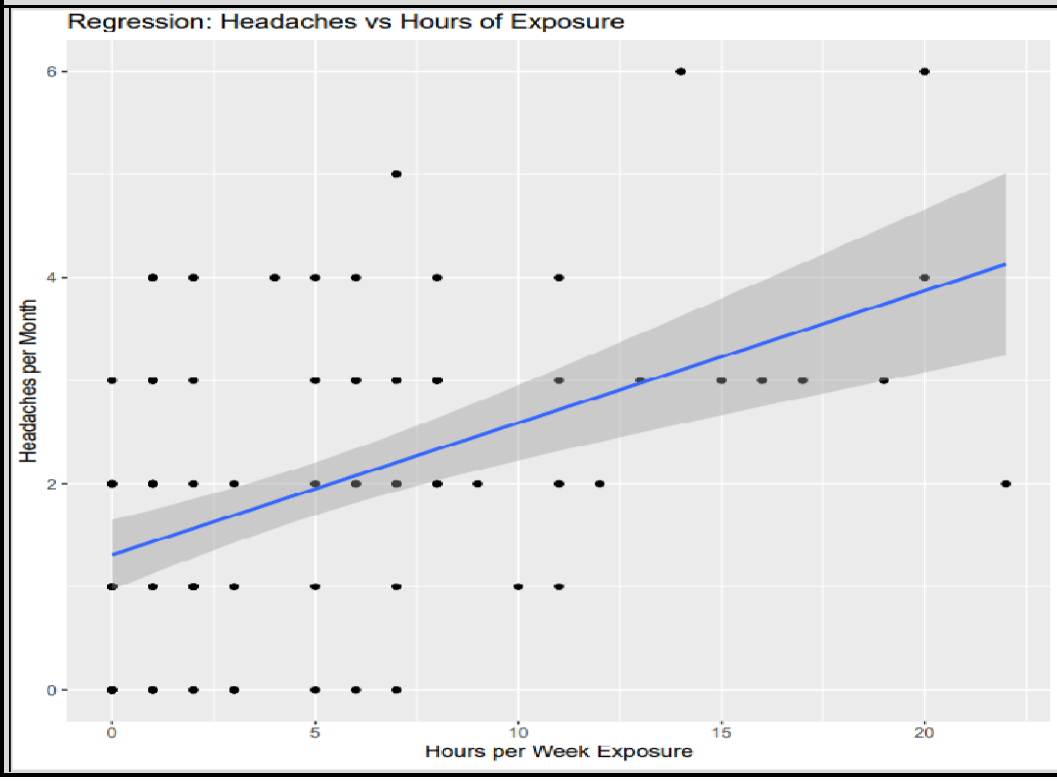
Figures

Analysis	Visualization Reference
R1 (Headaches vs. Exposure)	

R2a (Hours vs. Respiratory Symptoms)	

Analysis	Visualization Reference
R2b (Hours vs. Smoking Initiation)	
R3a (Physical Activity vs. Exposure)	

Analysis	Visualization Reference
R3b (Stress Level vs. Exposure)	
R4 (Parent Smoking vs. Started Smoking)	

Analysis	Visualization Reference
Regression (Headaches vs. Hours)	

Discussion

Summary of Statistical Findings and Interpretations

1. Exposure and Headaches (R1 & Regression) – Significant

Both the t-test and regression show a **strong positive relationship** between second-hand smoke exposure and monthly headaches.

- Exposed participants had **more headaches** (**t** ≈2.182 vs 0.957; **p ≈ 6.2×10⁻⁶**).
- Regression shows that each **additional hour of weekly SHS exposure increases headaches by 0.128** (**p ≈ 9.5×10⁻⁷**).

Reasoning: SHS exposure significantly increases headache frequency, indicating a direct physical health impact.

2. Exposure and Respiratory Symptoms (R2a)

The Kruskal-Wallis test found **strong differences** in exposure across symptom severity levels

(**p ≈ 6.47×10⁻⁵**).

Reasoning: More exposure is associated with more severe respiratory symptoms, suggesting a **dose-response relationship**.

3. Exposure and Smoking Initiation (R2b)

Participants who started smoking had **much higher SHS exposure** (9.313 vs 3.714 hours weekly; **$p \approx 0.004$**).

Reasoning: Higher SHS exposure may increase the likelihood of starting smoking, likely through normalization or increased access.

4. Non-Significant Findings (R3a, R3b, R4)

- **Physical Activity (R3a):** No difference between exposed and non-exposed (**$p = 0.8509$**).
  Exposure does not influence exercise habits.
- **Stress Level (R3b):** No difference in stress levels (**$p = 0.4165$**).
  SHS primarily affects physical health, not perceived stress.
- **Parent Smoking vs Starting Smoking (R4):** No association (**$p = 0.1244$**).
  Simply having a parent who smokes does **not** predict smoking initiation.

Personal Conclusion

Our research comes out with one unmistakable truth regarding secondhand smoke exposure: the dose is more important than the status.

We find strong evidence that the more hours someone spends around smoke, the more they suffer physically in terms of a significant rise in headaches and problems associated with the respiratory system. This is not just a correlation, our regression model confirms that every extra hour exposure per week adds about 0.13 to their predicted number of headaches.

But perhaps the most important finding is that the more exposure a person had previously, the far more likely they were to take up smoking themselves. Futhermore, having a parent who smokes wasn't sufficient to make a family member take up smoking themselves, what made this result was the intense, weekly hours of exposure.

Finally, though exposure unmistakably affects the body, we found no significant relation either to general stress levels or to physical activity. This suggests that the principal burden of SHS is physiological. The clear implication here is that public health efforts shouldn't just concentrate on 'if' someone is exposed but must focus on aggressively reducing the total number of exposure hours to protect both health and future behavioral choices.